2 Experimental design and procedures

We recruited children and adolescents to study the decision of an impartial observer who allocates rewards between two competitors. We follow the guidelines presented in Brocas and Carrillo (2020a) to address the methodological challenges that arise in studies with young participants.

Experiment. The experiment consists of two games programmed in 'oTree' (Chen et al., 2016). This study focuses on the first game, the third-party dictator. After a short break, participants played a two-person game with incomplete information, which we analyze in a separate paper (Alfonso et al., 2025).[3]

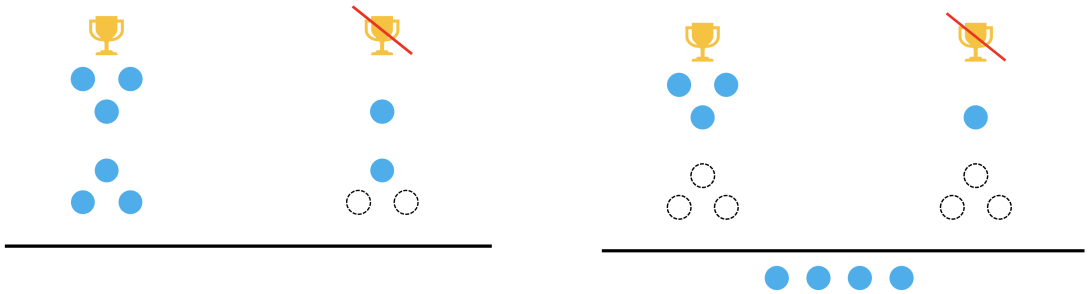
Timing. We first informed participants that they all would play an "observer task" followed by a "competition task". We emphasized that if a participant acted as the observer for two competitors, neither of those competitors would serve as their observer. In the observer task, participants allocated tokens between a future winner and a future loser of the competition task. In the competition task, they competed in pairs for a prize. We also informed them that feedback would be provided only at the end of the experiment. After the competition task, they were given the option to modify their decision in the observer task, with clear emphasis that no further modifications would be possible afterwards. These tasks were followed by the second game, an affective theory-of-mind task, and a Big 5 personality questionnaire (both described at the end of this section). The instructions for the observer and competition tasks are provided in Appendix A1.

Observer task. The impartial observer game used a visually intuitive interface (see Figure 1). Participants allocated tokens by dragging them between the winner (depicted by a trophy) and the loser (depicted by a crossed-out trophy). They could adjust the allocation freely until pressing OK to finalize their decision.

We explored two different environments for allocating tokens. First, a *redistribution* method, characterized by an initial allocation of (6,2) between the winner and the loser, with the observer able to redistribute 0, 1, or 2 tokens from the winner to the loser. This design is similar to the existing literature (Cappelen et al., 2022; Almås et al., 2020). The second is an *addition* design, where observers do not take tokens from the winner. Instead, the initial allocation is (3,1), and 4 additional tokens must be distributed between the two competitors, with at least one token allocated to each. Figures 1a and 1b show screenshots of both environments. Note that the three possible final allocations are identical in both environments: a "meritocratic" split (6,2) that maximizes the difference between winner and loser (while ensuring both receive a positive amount), a "compensatory" split (4,4) that equalizes final payoffs regardless of performance, and an "equitable" split (5,3) that

[3]The two games are distinct enough that we are not worried about cross-contamination. However, as a precautionary measure, we always conducted the two games in the same order, anonymously re-matching subjects between the two games, and did not disclose any result until the conclusion of the second game.

provides an intermediate option. However, we expect that the framing of the allocation method, take-from-the-rich (*redistribution*) vs. top-up (*addition*), may trigger different behavioral responses.



(a) *Redistribution*: 0, 1, or 2 tokens can be taken from winner and transferred to loser

(b) *Addition*: 4 tokens (bottom) must be allocated between winner and loser (with at least one for each)

Figure 1: Screenshots of the observer task: redistribution (left) v. addition (right)

Competition task. We explored three procedures for determining the winner of the competition, which encompass the three fundamental factors of performance in conventional economic models: *effort*, *talent* and *luck*.[4]

In the *effort* task, participants were randomly and anonymously paired to compete in an effort-intensive activity requiring patience and concentration. Following Gill and Prowse (2012), we employed a slider task. Each participants faced 39 non-aligned sliders, each starting at a random position between 0 and 100 (but never 50). The participant who successfully positioned the most sliders at exactly 50 within 90 seconds was the winner.[5] Prior to the competition, participants completed a short practice round where they adjusted three sliders to 50. Figure 2a provides a screenshot of the effort competition task, showing four sliders already set at 50 and a timer indicating 31 seconds remaining.

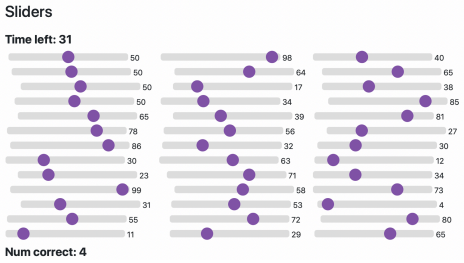
In the *talent* task, participants answered a series of 8 to 12 age- and culture-appropriate multiple-choice general knowledge questions, covering topics such as mathematics, biology, history, sports, and entertainment. Participants had, depending on age, 12 to 18 seconds to select one of four possible answers. The participant who answered the most questions correctly was declared the winner. Before the competition, participants completed a practice question to familiarize themselves with the format. Figure 2b displays a screenshot of the talent competition task (translated from the original Spanish version).

In the *luck* task, participants were simply informed that the computer would randomly

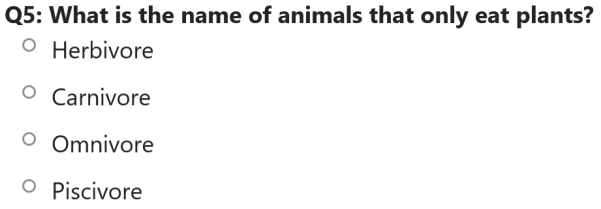
[4]Canonical models of performance are typically summarized by $y = f(e, \theta) + \eta$, where y = performance, e = effort, θ = ability, and η is a stochastic component.

[5]In case of a tie, the participant who reached that number of sliders first was the winner. Participants were never informed of the exact performance difference between winner and loser.

select a winner and a loser from each pair, with equal probability.



(a) *Effort*: participants had 90 seconds to position as many sliders as possible at 50



(b) *Talent*: participants answered age-appropriate, multiple choice general knowledge questions

Figure 2: Screenshots of the competition task: effort (left) v. talent (right)

Overall, the experiment consists of six variants, following a 3×2 between-subject design: three conditions for determining the winner (*effort*, *talent*, and *luck*) and two treatments for the allocation of tokens (*addition* and *redistribution*). Crucially, the design varies these two dimensions orthogonally. This one-factor-at-a-time structure allows us to attribute differences in outcomes to each dimension without confounding.

Population. We enrolled an unusually large sample for a lab-in-the-field experiment: 2,253 school-age students from low- to middle-income families, spanning grades 1 to 12 in the Spanish equivalent of the US education system. Participants came from four schools located in different cities across the southwestern region of Spain. All schools belong to Salesianos, a private network of Catholic schools funded by the Spanish government.[6] The experiment was integrated into the school's academic curriculum, with our Institutional Review Board (IRB) approving an opt-out procedure. Notably, no students opted out, eliminating the risk of selection bias. The only students who did not participate were those absent on the day of the experiment (5.7%). Treatment assignment was implemented at the class – or half-class – level (rather than at the individual level), while pairings within each session were anonymous and randomly determined. This approach reflects the realities of school-based implementation: students are scheduled and tested one class at a time, making class-level assignment the feasible and standard unit of randomization. Moreover, individual-level assignment is difficult to communicate to young children and could introduce confusion or demand effects. Table 1 summarizes the distribution of the 2,253 participants by grade, type of competition, and token allocation method. Of the 2,299 students who took part, 41 selected a gender category other than male or female and 5 did not report gender; these 46 observations are excluded from the analyses reported here, which use the binary Male control. Some further observations are excluded depending on

[6]The Salesianos educational network has 59 schools in Spain serving almost 40,000 students (see `https://www.salesianos.es/escuelas`).

the analysis: the regressions on age exclude 10 participants aged 225 months or more; in the *addition* environment, the software did not record the initial choice of 2 participants and the final choice of 14; and 8 participants have no DeRMET answers, 25 have incomplete P-Big 5 answers, and the task-perception questions were not recorded in two sessions (46 participants).

Grade	1	2	3	4	5	6	7	8	9	10	11	12	Total
addition / effort	28	30	32	31	20	24	54	27	23	50	23	0	342
addition / talent	22	26	25	23	52	47	30	29	51	25	31	24	385
addition / luck	22	22	22	24	24	50	26	80	54	46	29	45	444
redistribution / effort	19	28	23	25	45	22	54	52	31	28	23	34	384
redistribution / talent	32	8	22	21	49	43	27	26	31	25	29	20	333
redistribution / luck	15	22	33	33	25	26	62	29	28	43	23	26	365
Total	138	136	157	157	215	212	253	243	218	217	158	149	2253

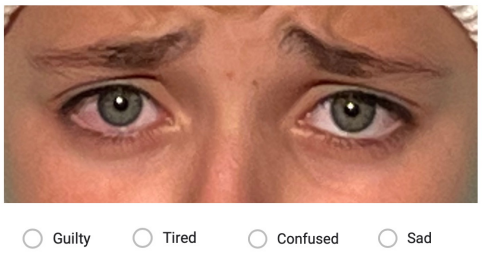
Table 1: Number of participants by grade, treatment, and condition

Payments. We identified a payment method that was uniform across all ages and highly valued by participants. All four schools in this sample have integrated cafeterias where students can purchase drinks, food and snacks at discounted prices. The cafeteria is very popular and widely accessed during recess and lunch, with items such as sandwiches and soft drinks highly and equally valued by students of all ages. To incentivize participants, we created our own currency, which could be spent on any item in the cafeteria. The currency had no expiration date, although most participants spent their endowments within a week. The exchange rate was 1 token = 1€.

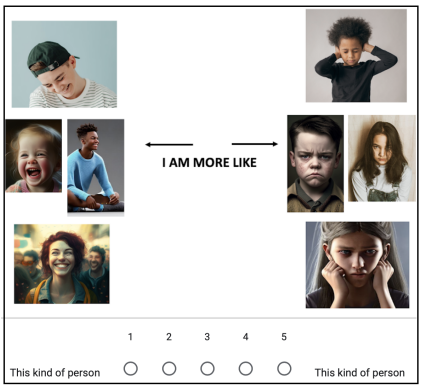
Developmental Read the Mind in the Eyes Task (DeRMET). Next, we introduced participants to a novel, incentivized affective theory-of-mind task to examine the link between social preferences and the ability to empathize with others. This task builds on the original "Read the Mind in the Eyes Task" (RMET), a psychological assessment developed by Baron-Cohen et al. (1997) to measure individuals' ability to attribute complex mental and emotional states to others. In that task, participants view images of a person's eyes and select the adjective that best describes the emotion being conveyed. The RMET was originally designed for adults, with emotions that reflect subtle nuances and are described using sophisticated language. The standard adaptation of the RMET to children, which consists in presenting a subset of the original images (Baron-Cohen et al., 2001), has been shown to not perform very well in young populations (van der Meulen et al., 2017). We therefore created an alternative RMET task, called DeRMET (Developmental RMET), specifically for children and adolescents. The task was tested, registered with the Open Science Framework (OSF), pre-validated with 43 teenagers and adults, and validated with the 2,253 observations collected in this experiment (see Appendix B7 for details). It is a

unique tool for researchers interested in studying the development of emotional intelligence and its association to decision-making processes.

Our adaptation features 16 images capturing the expressive eyes of a 12 years-old preteen who conveys 16 fundamental emotions, all described in simple language. For each image, we provide four possible answers, one of which most accurately represents the intended emotion. This modified version of the task offers three advantages for our population. First, sourcing all images from a single individual ensured greater consistency. Second, using a child as the model made the images more relatable for children and adolescents. Finally, focusing on simpler emotions and using straightforward language made the task appropriate for all ages, including the youngest participants.[7] Figure 3a shows one example image along with the four answer choices presented to participants (65% selected the correct answer, "sad").[8] Detailed instructions, including the full set of images, possible answers, correct responses, and the percentage of participants who answered each one correctly, are provided in Appendix A2.



(a) Developmental RMET



(b) Pictorial Big Five (neuroticism trait)

Figure 3: Examples of DeRMET and P-Big 5

Pictorial Big Five (P-Big 5). Next, we performed a novel version of the Big Five questionnaire, to study the joint evolution of personality and social preferences. A typical objective in psychology is to categorize individuals across the "Big 5" personality traits: openness, conscientiousness, extraversion, agreeableness, and neuroticism (Fiske, 1949; McCrae and Costa, 1987; Goldberg, 1993). Studies have employed a wide array of verbal questionnaires to examine those traits and how they relate to behavior and life

[7] Other authors have developed variations of the RMET task, or emotional intelligence tasks based on facial expressions (see e.g., Mattek et al. (2021); Pahnke et al. (2020)). To our knowledge, none are designed to study the developmental trajectory of affective theory-of-mind.

[8] An answer is labeled "correct" if it is *both* the emotion the preteen intended to convey in the image and the one chosen by the majority of participants. The two criteria coincide for all 16 images.

outcomes. These verbal questionnaires can pose significant challenges for young participants, particularly when subtle emotional states are involved. In studies with children, assessments often rely on variants completed by parents or caregivers. However, for the developmental trajectory of personality, it is crucial to employ a single assessment tool to ensure meaningful comparisons across various age groups.

To address these challenges and ensure consistent testing across individuals aged 6 to 18, we again developed a novel, child-friendly version of the Big 5 questionnaire, which we call the Pictorial Big 5 (P-Big 5). In this version, each trait is represented by a set of images illustrating opposite ends of the trait spectrum. Participants are shown a 5-point Likert scale and asked to select the option that best describes how they perceive themselves. The task was again tested, registered with OSF and pre-validated with 40 teenagers and adults who completed our P-Big 5 and a 25-question variant of the Five-Factor Model created for adolescents (Rogers and Glendon, 2018). We then validated the task with the 2,280 observations collected in this experiment (see Appendix B8 for details).

Our P-Big 5 questionnaire offers two advantages. Firstly, it does not rely on verbal comprehension, making it suitable for very young children and individuals from diverse linguistic backgrounds. Secondly, each question presents a natural choice between two polar ends, simplifying the decision-making process.[9] The primary limitation is that we offer just one question for each personality trait. This allows no space for cross-validation and focuses only on a specific facet of each dimension. Figure 3b presents the images employed for evaluating the "neuroticism" trait, and Appendix A3 provides the complete personality test, along with instructions and a comprehension example.

Questionnaires. At the end of the observer task, we asked participants in the effort and talent treatments to rate the difficulty of the competition task: 'Hard', 'Medium', 'Easy', or 'I don't know'. We also asked them to predict how was their performance compared to their competitor: 'Better', 'Equal', 'Worse', or 'I don't know'. These questions were not incentivized. Participants in the luck treatment were not asked to respond. We chose this timing intentionally to mitigate any potential demand effect that might prompt participants to alter their allocation in response to the task questions. At the end of the experiment we collected information regarding 'gender', 'date of birth', 'number of siblings' and 'favorite subject at school'.

Remarks. While we considered having each participant play only the observer or the competition task, we deemed this approach impractical, as it would have substantially reduced the number of observations for our primary task of interest—the observer task. In addition, our pilot study revealed that informing children that they would experience both roles enabled them to better understand the perspectives of both the winner and

[9]This contrasts with verbal questions that typically present only one extreme (e.g., "I Am interested in people") and ask the individual how much they agree with the statement. In our experience, children understand the question better if they can picture the two polar cases.

the loser, improving their comprehension of what it feels like to be in each position. By conducting the observer task before the competition task and then allowing participants to revise the allocation, we can document how decisions change after task participation. This sequence does not include a repeat-choice-only control condition, so changes may reflect experience, additional reflection, or repeated measurement; we therefore avoid assigning them a uniquely causal interpretation.